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ORIGINAL RESEARCH ARTICLE

Advancing Quality in the Evaluation, Surveillance, and Management of Aortic Stenosis: A Report From the AHA Target: AS Registry

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BACKGROUND: Undertreatment and delayed treatment of aortic stenosis (AS) are common, both of which are associated with increased mortality. Historically, assessment of quality for AS care has focused on peri- and postprocedural outcomes. The American Heart Association Target: AS registry is the first national registry to provide data and evaluate quality for upstream processes of care for patients with AS.

METHODS: Randomly selected patients from 2023 and 2024 with moderate or severe AS from 58 sites in the Target: AS registry were included. The 2 primary quality measures were (1) timely diagnosis (percentage of patients with an echocardiogram consistent with possible severe AS who had all assessments to clarify AS severity and symptoms within 30 days) and (2) timely treatment (percentage of patients with a Class I indication for aortic valve replacement treated within 90 days). Secondary measures included documentation of key echocardiographic parameters in the report, clinical recommendations in the echocardiogram report summary, multidisciplinary heart valve team evaluation, guideline-based performance of multimodality testing, and timely surveillance echocardiograms.

RESULTS: Among 8097 patients, 47% were women, 7% Black, 6% Hispanic, and 3% Asian. Timely diagnosis occurred in 54% in 2023, improving to 61% in 2024 (P=0.027) with gaps caused by lack of timely symptom assessment (23% [2023]; 16% [2024]), lack of stroke volume index (35% [2023]; 18% [2024]), or lack of timely multimodality testing (87% [2023]; 75% [2024]). Among those with a Class I indication for aortic valve replacement, timely treatment occurred in 82% (2023) versus 85% (2024) (P=NS). Multidisciplinary heart valve team evaluation occurred in 78% (2023) versus 84% (2024) (P=0.008). Key findings in the echocardiography report were documented in 83% (2023) and 85% (2024) (P=NS), but a clinical recommendation was included in <10% of the time. Timely surveillance echocardiograms for moderate (2 years) and severe AS (1 year) were not performed in ~40% of patients.

CONCLUSIONS: The national American Heart Association Target: AS registry demonstrates opportunities for improvement in timely diagnosis, surveillance, and treatment for patients with AS at participating centers. By providing an infrastructure to measure performance and identify best practices, the Target: AS registry has the potential to elevate and optimize care and patient outcomes.

Key Words: aortic valve stenosis ■ aortic valve replacement ■ echocardiography ■ quality improvement ■ registries

Over the last 2 decades, numerous randomized clinical trials have demonstrated the safety and efficacy of transcatheter aortic valve replacement as an alternative to surgery for the treatment of severe aortic stenosis (AS) at all levels of surgical risk.[1–3] This has expanded treatment for higher risk patients

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Clinical Perspective

What Is New?

- This is the first large-scale report from the American Heart Association Target: Aortic Stenosis (AS) registry, which provides unique insights about processes of care for AS upstream of aortic valve replacement.
- Timely diagnosis occurred just over half the time, and two-thirds of those with discordant echocardiogram measurements lacked guideline-recommended testing to clarify AS severity.
- More than 80% of patients with a known Class I indication for aortic valve replacement were treated within 90 days.
- Timely follow-up surveillance echocardiograms were missing 40% of the time for patients with known moderate or severe AS.

What Are the Clinical Implications?

- There is a need to improve performance on these quality measures and reduce variability across sites.
- Systems-based best practices need to be identified, implemented, and scaled to improve the quality of care and optimize outcomes for patients with AS.

Nonstandard Abbreviations and Acronyms

AHA	American Heart Association
AS	aortic stenosis
AVA	aortic valve area
AVR	aortic valve replacement
LVEF	left ventricular ejection fraction
MDT	multidisciplinary heart valve team
QI	quality improvement
SVI	stroke volume index

who were either poor candidates for or who did not want to undergo surgery.[4,5] Technology- and device-related improvements alongside standardized best practices for preprocedural evaluation and intraprocedural techniques have been associated with a steady decline in risk and complications for both transcatheter and surgical therapies.[6,7] Despite these advances, many patients with severe AS are still not offered aortic valve replacement (AVR) as a result of a variety of factors, or treatment is delayed. Delayed treatment has been associated with death on a waiting list or urgent AVR, often in the setting of advanced heart failure.[8–11] In response to these consequential care gaps, the American College of Cardiology/American Heart Association (AHA) recently published a performance metric with the goal of having eligible patients with symptomatic severe AS undergo AVR within 90 days of their diagnosis.[8,9,12–14]

To support improvement in the systems of care contributing to the suboptimal evaluation and treatment of patients with AS, the AHA launched the Target: AS initiative in 2020.[15] Whereas the Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy Registry and Society of Thoracic Surgeons Adult Cardiac Surgery Database focus on intraprocedural, in-hospital, and postdischarge care after AVR, Target: AS was created to identify and address quality improvement (QI) opportunities upstream of the procedure. The Target: AS registry is a voluntary, hospital-based QI program that was initiated to improve the care of patients with structural heart disease by promoting consistent adherence to treatment guidelines.[15] The initial 3-year pilot of Target: AS included 15 sites that were tasked with tracking care for a subset of patients with severe or moderate AS with a provisional set of quality metrics evaluated.

The Target: AS initiative has rapidly expanded and evolved into a mature and scalable registry. The network has grown to include 80 sites that vary in geography, size, and setting (community versus academic). Measures were carefully selected and tested and now 2 primary and several supportive secondary QI measures exist. In this report, we provide data on these quality measures across the Target: AS network, highlighting gaps in care and related opportunities to improve the evaluation, management, and treatment of patients with AS.

METHODS

The authors declare that all supporting data are available within the article and its Supplemental Material.

Site Selection

Participation in Target: AS is broadly inclusive with open invitations to hospitals to participate. Hospitals participating in the Target: AS registry use an interactive, web-based tool to enter and transmit clinical information to a central registry. The data are de-identified, centrally aggregated, and monitored for quality.

Patients and Sampling

This observational cohort study from the Target: AS registry includes adult patients (>18 years old) with moderate or severe AS and no history of AVR in the registry. Patients were eligible for inclusion if they had at least 1 echocardiogram at the site. Moderate AS was defined by an aortic valve area (AVA) of 1.01 to 1.50 cm^2 and peak transvalvular velocity of 2.5 to 3.9 m/s, whereas severe (or potentially severe) AS was defined by an AVA ≤1.0 cm^2 or peak transvalvular velocity ≥4 m/s. From among all eligible patients identified at their site through an unbiased query of their echocardiography database, sites sampled a random 5% of patients with moderate AS and a random 15% of patients with severe AS for registry enrollment (see Supplemental Methods for further detail). This sampling plan was instituted so as to gather a representative sample without overburdening sites with data entry. Low volume sites were asked to abstract at least 40 patients with severe AS and 20 patients with moderate AS annually; in contrast, high volume sites had a maximum yearly threshold of 150 patients with severe AS and 75 patients with moderate AS.

Registry Data Captured

Registry data included patient characteristics, demographics, and comorbidities. Data from the initial qualifying echocardiogram were also included, along with any subsequent surveillance echocardiograms performed before AVR. For any patient with definite (or potentially) severe AS, additional data related to clinical encounters (eg, clinic visit, hospitalization) and diagnostic testing (eg, computed tomography scans, dobutamine stress echocardiogram, treadmill stress testing, or cardiac catheterization) were collected. Last, information related to multidisciplinary heart valve team (MDT) evaluation and decision-making about treatment were collected. Complete case report forms are included in the Supplemental Material. Data collection and analysis adhered to all relevant ethical guidelines, including maintaining patient confidentiality and data security. Hospitals participating in Target: AS were given resource documents to guide program participation, including patient entry criteria, data element definitions, and measure descriptions. Each participating hospital either received approval to collect patient-level data without patient consent (under the common rule) or a waiver of authorization and exemption from subsequent review by their institutional review board. Advarra, the institutional review board for the American Heart Association, determined that this study was exempt from institutional review board oversight. IQVIA (Cambridge, MA) served as the data collection and coordinating center.

Primary Quality Measures

The 2 primary quality measures were timely diagnosis and timely treatment. Timely diagnosis was defined as the percentage of patients who had all required diagnostic testing and clinical evaluations to help determine whether a Class I indication for AVR existed within 30 days of an echocardiogram that demonstrated definite (or potentially) severe AS (AVA ≤1.0 cm^2 or mean gradient ≥40 mm Hg or peak gradient ≥64 mm Hg or peak velocity ≥4 m/s).[16] These required elements (for "defect-free" timely diagnosis) included (1) assessment of symptoms; (2) a left ventricular ejection fraction (LVEF) in the echocardiogram report, (3) stroke volume index (SVI) in the echocardiogram report (when transvalvular gradients were lower and the LVEF was ≥50%), and (4) multimodality diagnostic testing with an aortic valve calcium score, dobutamine stress echocardiogram, or invasive hemodynamics to clarify AS severity (when transvalvular gradients were lower in a patient with a LVEF <50% or when SVI was <35 mL/m^2 along with a LVEF ≥50%). Exclusions for this measure included reports of echocardiograms performed on the same date or after the AVR procedure. Exceptions for this measure included patients who died or had a documented reason for discontinuing care within 30 days of the echocardiogram of interest.

Timely treatment was defined as the percent of patients treated with AVR within 90 days of being deemed to have a Class I indication for AVR.[16] Exclusions for this measure included patients who had AVR performed during a previous measurement period or whose AVR was performed before the earliest date a Class I Indication for AVR was identified. Exceptions for this measure included patients who had a documented reason for not referring for AVR consideration (eg, AS deemed nonsevere based on aortic valve calcification or dobutamine stress echocardiogram, a life-limiting comorbidity with a life expectancy <1 year, or presence of significant dementia), patients whose treatment plan was determined to be definitively nonprocedural (eg, patient refused AVR despite its recommendation, patient was referred to palliative care, or the multidisciplinary team determined AVR would be futile), patients who died or had a documented reason for discontinuing treatment within 90 days of their earliest Class I indication, or if an accepted reason for delay in AVR was recorded (eg, dental work or tooth extraction, incidental finding on a computed tomography scan that required further evaluation, insurance denial, patient preference for delayed treatment, or management of another competing medical problem).

Secondary Quality Measures

The secondary quality measures included the percentage of patients undergoing AVR who were seen by the MDT (evaluation by both a cardiologist and cardiac surgeon on the MDT before the AVR); the percentage of echocardiogram reports with complete information (defined as an AVA, mean gradient, peak gradient [or velocity], and LVEF anywhere in the report and a qualitative description of AS severity in the summary/conclusion); a clinical recommendation in the echocardiogram report summary; and the percent of echocardiograms with timely follow-up echocardiographic surveillance (ie, within 12 months for severe AS and within 24 months for moderate AS). For the surveillance echocardiogram metrics, death during the surveillance period is an exclusion (that is, the patient is removed from both the numerator and the denominator).

Statistical Approach

We compared 2023 and 2024 performance for the quality measures. Because of rapid growth in the number of sites that could influence comparisons, we compared 2023 with 2024 data for 3 different groups/cohorts: (1) all sites with a minimum of 1 patient eligible for any measure in either year (n=58); (2) sites with a minimum of 40 patients entered in the registry for both 2023 and 2024 (n=26); and (3) sites who had joined Target: AS before 2023 (n=12). Data were summarized descriptively using counts and percentages. Performance rates in the manuscript and bar charts were derived by summing the total numerators and dividing by the total denominators per measure, year, and site cohort. Box-and-whisker plots depict medians and interquartile ranges. The lower and upper whiskers were defined as the 2.5th and 97.5th percentiles, respectively. For the "all sites" cohort, performance rates for 2023 and 2024 were compared among paired sites using a 2-sided Wilcoxon signed-rank test. For the "volume-qualifying sites" and "long-term sites" cohorts, all sites were by definition paired, so performance rates for 2023 and 2024 were compared using a 2-sided Wilcoxon signed-rank test. In addition, to account for clustering of patients within sites and encounters within patients, we fit unadjusted and adjusted generalized linear mixed-effects models with random intercepts for both site and patient to compare encounter-level timely diagnosis between 2023 and 2024 among volume-qualifying sites. Sites with <20 encounters per year were excluded. The adjusted model included sex, age, heart failure, diabetes, coronary artery disease, other valvular heart diseases, dyslipidemia, atrial fibrillation, chronic kidney disease, chronic obstructive pulmonary disease, and previous myocardial infarction as covariates. Results were reported as odds ratios with 95% CIs.

Descriptive statistics and bar charts were calculated in Microsoft Excel. Box-and-whisker plots were calculated using GraphPad Prism 11.0.2 for macOS. The Wilcoxon signed-rank tests were conducted using Python version 3.11.3 and the scipy (1.10.1) package. Mixed-effects models were fit in R version 4.4.1 using the lme4 (1.1-37) package. All statistical analyses were conducted by the American Heart Association with $P<0.05$ considered statistically significant.

RESULTS

Site and Patient Characteristics

There are 58 sites represented in this report with diverse characteristics based on geography, hospital type, bed size, and teaching status (Table S1; Figure S1). The age, race, and ethnicity of 8097 patients entered into the registry are shown in Table 1. There was a broad range of ages included with 15% <65 years of age and 9% >90 years of age. Approximately 47% of patients were

Table 1. Patient Characteristics

Characteristics	All active sites with patients in 2023 or 2024 (58 sites) (8097 patients)	Volume-qualifying sites with a minimum of 40 patients in both 2023 and 2024 (26 sites) (5610 patients)	Long-term sites that joined before 2023 (12 sites) (3378 patients)
Women	3800 (47%)	2596 (46%)	1522 (45%)
Median age	76.5	76.0	76.0
Age groups, y			
18–49	206 (3%)	168 (3%)	100 (3%)
50–64	991 (12%)	714 (13%)	419 (12%)
65–79	3685 (46%)	2551 (45%)	1584 (47%)
80–89	2473 (31%)	1682 (30%)	993 (29%)
≥90	742 (9%)	495 (9%)	282 (8%)
Race/ethnicity			
Non-Hispanic White	6451 (80%)	4379 (78%)	2632 (78%)
Non-Hispanic Black	554 (7%)	421 (8%)	256 (8%)
Hispanic White	312 (4%)	179 (3%)	99 (3%)
Hispanic Black	5 (0%)	4 (0%)	2 (0%)
Hispanic, other race or no race	154 (2%)	118 (2%)	78 (2%)
Asian	242 (3%)	197 (4%)	97 (3%)
American Indian or Alaska Native	35 (0%)	25 (0%)	13 (0%)
Native Hawaiian or Pacific Islander	23 (0%)	23 (0%)	6 (0%)
Middle Eastern or North African	2 (0%)	0 (0%)	0 (0%)
Unable to determine	329 (4%)	270 (5%)	194 (6%)
Race and ethnicity are blank	5 (0%)	5 (0%)	5 (0%)
Comorbidities			
Heart failure	2908 (36%)	2023 (36%)	1354 (40%)
Diabetes	2756 (34%)	1900 (34%)	1256 (37%)
Coronary artery disease	2730 (34%)	1847 (33%)	1284 (38%)
Multiple valvular heart disease	2002 (25%)	1770 (32%)	1261 (37%)
Dyslipidemia	1941 (24%)	1059 (19%)	804 (24%)
Atrial fibrillation	1941 (24%)	1249 (22%)	772 (23%)
Chronic kidney disease	1881 (23%)	1293 (23%)	840 (25%)
Chronic obstructive pulmonary disease	1034 (13%)	719 (13%)	440 (13%)
Myocardial infarction	858 (11%)	632 (11%)	402 (12%)
Peripheral artery disease	790 (10%)	562 (10%)	411 (12%)

women, 7% were Black, 6% were Hispanic, and 3% were Asian.

Diagnostic Testing and Clinical Encounters

Over 32 000 clinical or diagnostic testing encounters were captured through 2023 and 2024, including almost 10 917 resting echocardiograms and 9284 outpatient clinical encounters (Table S2). Completeness of echocardiography reports with respect to key AS-related variables is also shown in Table S2. The most commonly missing parameter was SVI (missing 41% of the time in 2023 and 23% in 2024).

Timely Diagnosis

Timely diagnosis occurred in approximately 55% of patients in 2023 across the different cohorts (Figure 1). This rate improved by 7% to 10% in 2024 across the different cohorts ($P<0.05$ for all comparisons). In both years and across all the cohorts, there was marked variability in site performance for this measure (Figure 1). In a generalized linear mixed-effects model accounting for clustering of patients within sites and encounters within patients, timely diagnosis was higher in 2024 than in 2023 (odds ratio, 1.54 [95% CI, 1.33–1.79]; $P<0.001$), and this association was unchanged after adjustment for sex, age, and comorbidities (odds ratio, 1.52 [95% CI, 1.31–1.76]; $P<0.001$). Figure 2 and Table 2 provide more detailed information about measure performance. In 2023 or 2024, there were 5821 patients with an echocardiogram with definite (or possibly) severe AS across all sites. Of those, 3126 (54%) had high-gradient severe AS, and 2695 (46%) had discordant measurements (AVA ≤1.0 cm^2 with mean gradient <40 mm Hg and peak velocity <4.0 m/s). Of those with discordant measurements, 1707 (63%) were missing information necessary to further clarify AS severity, which was almost entirely a result of missing stroke volume index (when needed) (representing one-third of the missingness) or missing/delayed multimodality testing (representing two-thirds of the missingness). Among the one-third of patients (n=988) with discordant measurements who did have data to assess AS severity, 545 (55%) had LVEF ≥50% and normal SVI, which excluded severe AS based on current guideline definitions.[16] Of the remaining 443 patients, 254 (57%) were found to have severe AS after multimodality testing. Patients with confirmed severe AS lacked a documented assessment of symptoms 6% (196 of 3380) of the time. Of the 3184 patients with confirmed severe AS and symptom assessment available, 2664 (84%) had a Class I indication for AVR, and 520 (16%) did not.

Figure 3 shows component performance for the timely diagnosis measure. Timely symptom assessment was performed 77-82% of the time in 2023 (range across different cohorts) with an improvement in the rate in 2024. When required, SVI was included less than two-thirds of the time in the echocardiography report in 2023, with substantial improvement in 2024. There was marked variability in reporting SVI across sites in 2023 that lessened somewhat in 2024. When warranted, timely multimodal testing was only performed 11% to 13% of the time in 2023, which improved to one-quarter of the time in 2024. Again, marked variability in site performance was found for timely multimodal testing.

Timely Treatment

Among patients with a Class I indication for AVR, 81% to 84% were treated with AVR within 90 days during 2023 (Figure 1). Across cohorts, this numerically increased 0.3% to 3.8% in 2024. There was also substantial variability in performance across sites for this measure, although less variability was found among sites that contributed more patients to the registry or had participated in the registry longer (Figure 1). Reasons for and frequency of exclusions from this measure are shown in Table S3.

Treatment

Among patients with a Class I indication for AVR, approximately 6% had a qualifying reason for nonprocedural treatment (eg, MDT determined AVR would be futile or patient refused), 58% were treated with transcatheter aortic valve replacement, 18% were treated with SAVR, and 19% did not have any treatment documented or reason for pursuing nonprocedural treatment (Table 3). Among those with confirmed severe AS but no Class I indication for AVR (primarily because of the lack of symptoms), 26% were treated with transcatheter aortic valve replacement, 8% were treated with SAVR, 9% had a definitive plan for nonprocedural treatment based on a qualifying reason, and 58% did not have any treatment documented (Table 3).

Secondary Quality Measures

Among those treated with AVR, 76% to 79% were evaluated by the MDT before AVR in 2023 (Figure 4). This increased by 5% to 7% across different cohorts in 2024. Approximately 83% of echocardiography reports had the required information relevant to AS diagnosis in 2023, with a modest numerical (but nonsignificant) increase in 2024 (Figure 4). A clinical recommendation was rarely (1 of 10) included in the echocardiogram report summary statement (Figure 4). Timely follow-up surveillance echocardiograms for severe AS were performed <50% of the time in 2023 with a numerical increase of 12% in 2024; there was substantial variability in site

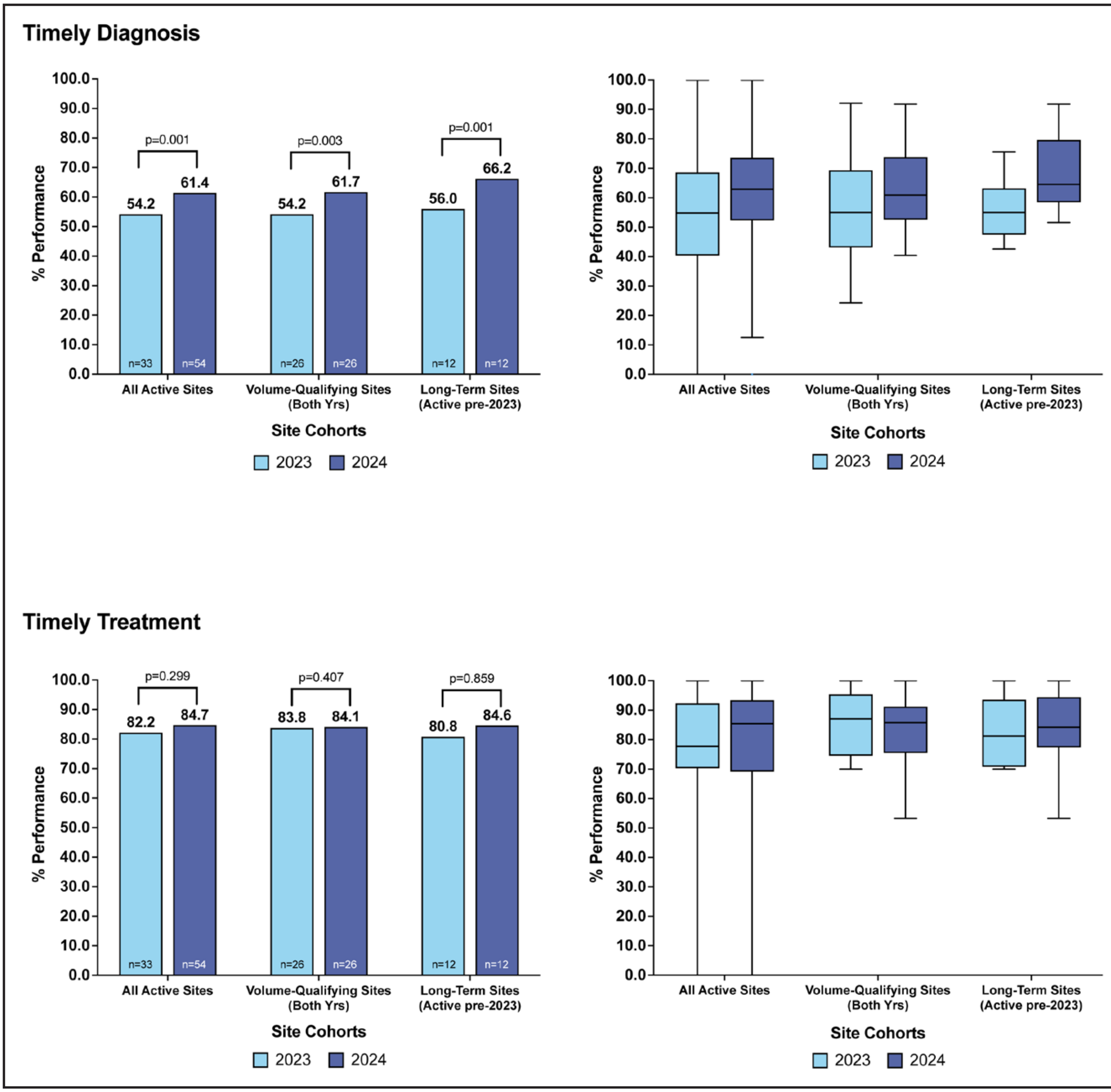

Figure 1. Timely diagnosis and timely treatment (primary quality measures of Target: Aortic Stenosis).

performance in 2023 that lessened in 2024 (Figure 5). For moderate AS, timely follow-up surveillance echocardiograms were performed 63% of the time in both 2023 and 2024 with greater site variability in 2024 (Figure 5).

Symptom Status by Severity of AS

Table 3 shows the symptom status (symptomatic, asymptomatic, or unknown) for the 3 groups of patients: (1) patients with definite (or potentially) severe AS based on echocardiography, (2) patients with confirmed severe AS, and (3) patients with concordant moderate AS or discordant measurements who were shown to have nonsevere AS. Among patients with moderate AS, approximately half had symptoms that might be related to AS.

DISCUSSION

The diagnosis and management of AS are inherently multidisciplinary, requiring coordinated systems of care among primary care, general cardiology, cardiovascular imaging (echocardiography and cardiac CT), interventional cardiology, cardiac surgery, and other select members of the heart team. Against a background of accumulating data pointing to adverse health consequences related to under- and delayed treatment of patients with severe AS, the AHA Target: AS quality improvement program was

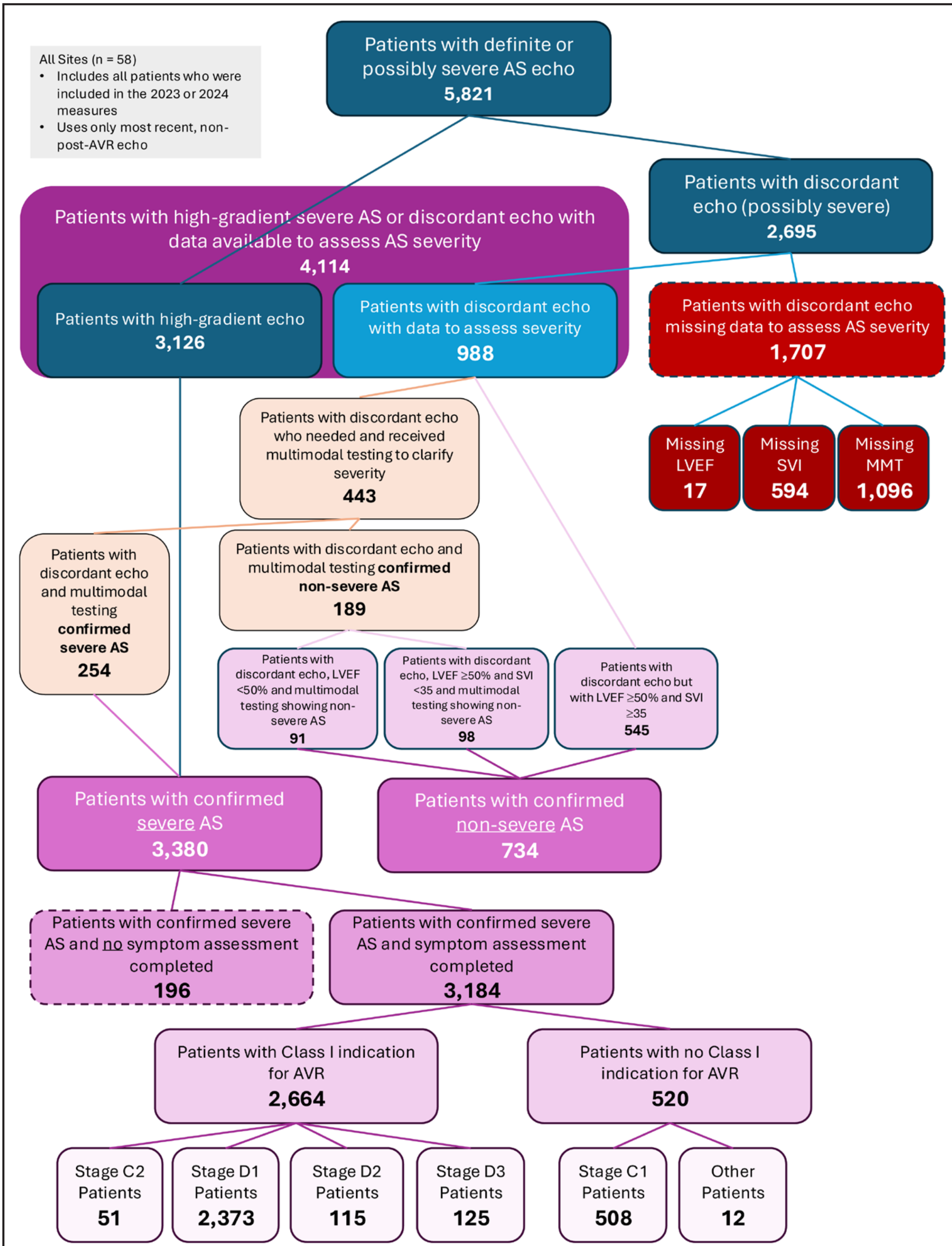

Figure 2. Timely diagnosis–AS severity and missing data.
Stages C and D are as defined in the American College of Cardiology/American Heart Association Valvular Heart Disease Guidelines.[16] AS indicates aortic stenosis; AVR, aortic valve replacement; echo, echocardiogram; LVEF, left ventricular ejection fraction; MMT, multimodal testing; and SVI, stroke volume index.

Table 2. AS Severity, Testing Completed, and AS Stages

Patient subgroups	All active sites with patients in 2023 or 2024 (58 sites)	Sites with a minimum of 40 patients in both 2023 and 2024 (26 sites)	Long-term sites (12 sites)
Patients with resting echocardiogram	8006	5540	3325
Patients with definite or possible severe AS echocardiogram*	5821 of 8006 (73%)	3999 of 5540 (72%)	2422 of 3325 (73%)
Patients with high-gradient echocardiogram†	3126 of 5821 (54%)	2206 of 3999 (55%)	1374 of 2422 (57%)
Patients with discordant echocardiogram (possibly severe)‡	2695 of 5821 (46%)	1793 of 3999 (45%)	1048 of 2422 (43%)
Patients with discordant echocardiogram missing data to assess severity	1707 of 2695 (63%)	1214 of 1793 (68%)	710 of 1048 (68%)
Patients without LVEF when needed	17 of 1707 (1%)	6 of 1214 (0%)	2 of 710 (0%)
Patients without SVI when needed (LVEF ≥50%)	594 of 1707 (35%)	523 of 1214 (43%)	359 of 710 (50%)
Patients without multimodal testing when needed§	1096 of 1707 (64%)	685 of 1214 (56%)	349 of 710 (49%)
Patients with discordant echocardiogram with data to assess severity	988 of 2695 (37%)	579 of 1793 (32%)	338 of 1048 (32%)
Patients with discordant echocardiogram not needing multimodal testing to clarify severity‖	545 of 988 (55%)	313 of 579 (54%)	176 of 338 (52%)
Patients with discordant echocardiogram who needed and received multimodal testing to clarify severity	443 of 988 (45%)	266 of 579 (46%)	162 of 338 (48%)
Multimodal testing confirmed severe AS	254 of 443 (57%)	144 of 266 (54%)	87 of 162 (54%)
Multimodal testing confirmed nonsevere AS	189 of 443 (43%)	122 of 266 (46%)	75 of 162 (46%)
Patients with concordant moderate AS echocardiogram¶	1422 of 8006 (18%)	906 of 5540 (16%)	600 of 3325 (18%)
Patients with mild AS echocardiogram#	763 of 8006 (10%)	635 of 5540 (11%)	303 of 3325 (9%)
Patients with high-gradient severe AS or discordant echocardiogram with data available to assess AS severity	4114	2785	1712
Patients with confirmed nonsevere AS	734 of 4114 (18%)	435 of 2785 (16%)	251 of 1712 (15%)
Patients with discordant echocardiogram but with LVEF ≥50% and SVI ≥35	545 of 734 (74%)	313 of 435 (72%)	176 of 251 (71%)
Patients with discordant echocardiogram, LVEF <50%, and multimodal testing showing nonsevere AS	91 of 734 (12%)	54 of 435 (12%)	29 of 251 (11%)
Patients with discordant echocardiogram, LVEF ≥50%, SVI <35, and multimodal testing showing nonsevere AS	98 of 734 (13%)	68 of 435 (16%)	46 of 251 (18%)
Patients with confirmed severe AS**	3380 of 4114 (82%)	2350 of 2785 (84%)	1461 of 1712 (85%)
Patients with confirmed severe AS and symptom assessment completed	3184 of 3380 (94%)	2215 of 2350 (94%)	1404 of 1461 (96%)
Patients with no Class I indication for AVR	520 of 3184 (16%)	333 of 2215 (15%)	226 of 1404 (16%)
Stage C1	508 of 520 (98%)	327 of 333 (98%)	223 of 226 (99%)
Other††	12 of 520 (2%)	6 of 333 (2%)	3 of 226 (1%)
Patients with a Class I indication for AVR	2664 of 3184 (84%)	1882 of 2215 (85%)	1178 of 1404 (84%)
Stage C2	51 of 2664 (2%)	28 of 1882 (1%)	13 of 1178 (1%)
Stage D1	2373 of 2664 (89%)	1718 of 1882 (91%)	1083 of 1178 (92%)
Stage D2	115 of 2664 (4%)	60 of 1882 (3%)	32 of 1178 (3%)
Stage D3	125 of 2664 (5%)	76 of 1882 (4%)	50 of 1178 (4%)
Patients with confirmed severe AS and no symptom assessment completed	196 of 3380 (6%)	135 of 2350 (6%)	57 of 1461 (4%)

Stages C and D are as defined in the American College of Cardiology/American Heart Association Valvular Heart Disease Guidelines.[16] AS indicates aortic stenosis; AVA, aortic valve area; AVR, aortic valve replacement; LVEF, left ventricular ejection fraction; and SVI, stroke volume index.

*Severity is based on the patient's most recent pre-AVR echocardiogram or, if no AVR, most recent echocardiogram.

†High-gradient echocardiogram = peak velocity ≥4.0 m/s or peak gradient ≥64 mm Hg or mean gradient ≥40 mm Hg.

‡Discordant echocardiogram (possibly severe) = AVA ≤1.0 cm^2 and (peak velocity <4.0 m/s or peak gradient <64 mm Hg or mean gradient <40 mm Hg) and no high-gradient (peak velocity ≥4.0 m/s or peak gradient ≥64 mm Hg or mean gradient ≥40 mm Hg) values.

§Discordant echocardiogram and LVEF <50% or discordant echocardiogram and LVEF ≥50% and SVI <35.

‖Discordant echocardiogram and LVEF ≥50% and SVI ≥35.

¶No high gradients, AVA >1.0 cm^2 to ≤1.5 cm^2, and (peak velocity ≥2.5 to <4.0 m/s) or (peak gradient ≥25 to <64) or (mean gradient ≥20 to <40).

#No high gradients, discordant (potentially severe) AS, or moderate AS.

**High-gradient AS or (discordant echocardiogram with LVEF <50% and multimodal testing confirmed severe) or (discordant echocardiogram with LVEF ≥ 50% and SVI <35 and multimodal testing is severe).

††(Asymptomatic and high-gradient AS and no LVEF) or (asymptomatic and discordant echocardiogram with LVEF <50% and multimodal testing indicated severe AS) or (asymptomatic and discordant echocardiogram with LVEF ≥50% and SVI <35 and multimodal testing indicated severe AS).

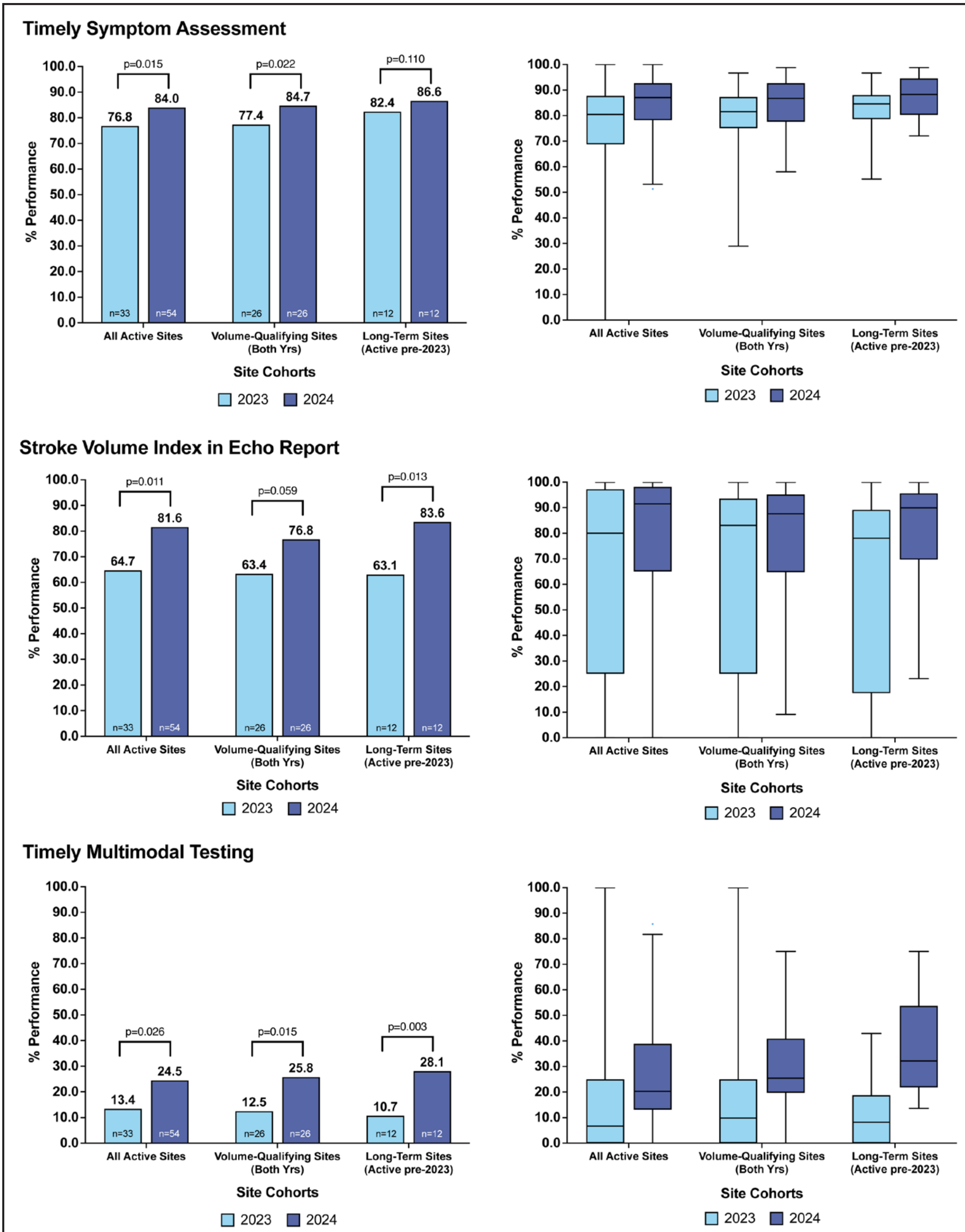

Figure 3. Components of timely diagnosis measure.
The components of the timely diagnosis measure are shown. The inclusion of stroke volume index in the echocardiography report and performance of multimodal testing is assessed only when those measures are warranted based on the aortic valve area, transvalvular gradients, and LVEF. Note that inclusion of LVEF in the echocardiography report is also a component of the timely diagnosis measure, but it was present 99% of the time (Figure S2). Echo indicates echocardiogram; and LVEF, left ventricular ejection fraction.

Table 3. Treatment Types and Symptom Status

Patient subgroups	All active sites with patients in 2023 or 2024 (58 sites)	Sites with a minimum of 40 patients in both 2023 and 2024 (26 sites)	Long-term sites (12 sites)
Patients with confirmed Class I indication for AVR	2664	1882	1178
Received TAVI	1536 (58%)	1059 (56%)	668 (57%)
Received SAVR	467 (18%)	363 (19%)	242 (21%)
Received qualifying nonprocedural treatment*	166 (6%)	117 (6%)	65 (6%)
No treatment documented	495 (19%)	343 (18%)	203 (17%)
Patients with no Class I indication for AVR (98% were stage C1)	520	333	226
Received TAVI	135 (26%)	94 (28%)	67 (30%)
Received SAVR	40 (8%)	30 (9%)	25 (11%)
Received qualifying nonprocedural treatment	49 (9%)	33 (10%)	17 (8%)
No treatment documented	296 (57%)	176 (53%)	117 (52%)
Patients with definite or possible severe AS echo	5821	3999	2422
Symptomatic†	4081 (70%)	2863 (72%)	1785 (74%)
Asymptomatic‡	1178 (20%)	745 (19%)	439 (18%)
Unknown	562 (10%)	391 (10%)	198 (8%)
Patients with confirmed severe AS	3380	2350	1461
Symptomatic	2613 (77%)	1854 (79%)	1165 (80%)
Asymptomatic	571 (17%)	361 (15%)	239 (16%)
Unknown	196 (6%)	135 (6%)	57 (4%)
Patients with confirmed nonsevere AS or moderate AS	2156	1341	851
Symptomatic	1036 (48%)	710 (53%)	472 (55%)
Asymptomatic	430 (20%)	288 (21%)	191 (22%)
Unknown	690 (32%)	343 (26%)	188 (22%)

AS indicates aortic stenosis; AVR, aortic valve replacement; SAVR, surgical aortic valve replacement; and TAVI, transcatheter aortic valve implantation.

*Definitive treatment decision was nonprocedural because the multidisciplinary team determined AVR to be futile, the patient refused AVR, or treatment is palliative care.

†Exists (outpatient physical assessment where symptoms were "shortness of breath," "exertional dyspnea," "decreased exercise tolerance," "angina/chest pain," "presyncope," or "syncope") OR exists (inpatient reason for admission where symptoms were "shortness of breath/heart failure," "chest pain," or "syncope or presyncope") OR exists (exercise stress test where symptoms were "chest pain," "presyncope," "syncope," "dyspnea or fatigue that required stopping," "shortness of breath," or "reduced activity level").

‡Never symptomatic AND exists (outpatient physical assessment where "symptoms not assessed or unknown" <> TRUE (not equal to), and symptoms were "none," "heart murmur," "reduced activity level," "other," or "symptoms assessed but absent") OR exists (inpatient reason for admission where symptoms were "none," "atrial arrhythmia," or "myocardial infarction") OR exists (exercise stress test where symptoms were "none" or "other" or "heart palpitations").

developed to highlight and address gaps in the management and treatment of patients with AS by tracking adherence to guideline-recommended management, outlining best practices to decrease process and care gaps, and reinforcing a multidisciplinary approach. Growth and evolution of the initiative included an expanded number of sites, refinement of the timely treatment metric, and addition of a timely diagnosis metric (Figure 6). To our knowledge, this is the first such patient registry to provide insight into these processes at scale with thousands of patients across sites with a diversity of practice settings and geography.

In this report from the Target: AS registry, there are 5 main findings. First, timely diagnosis was achieved just over half the time. Among patients not meeting the timely diagnosis metric, two-thirds of those with discordant echocardiogram measurements lacked guideline-recommended clarifying testing (dobutamine stress echocardiogram or aortic valve calcium score) resulting in indeterminate AS severity. This points to large numbers of patients with potentially severe AS not being correctly identified, which may contribute to delayed treatment or undertreatment with AVR. Second, timely diagnosis and appropriate testing may have been limited by absence of SVI (missing in one-fourth to one-third of reports when needed) and lack of clinical recommendations in the echocardiogram report (missing 90% of the time), which may help direct ordering clinicians to follow up with clarifying multimodal testing. Third, among patients with necessary assessments to determine AS severity and symptom status, those with a Class I indication for AVR were treated within 90 days at a fairly high rate (>80%). Fourth, there was improvement in timely diagnosis from 2023 to 2024, potentially associated with QI reporting in the registry. Despite this, substantial variability was observed between sites with regard to timely diagnosis

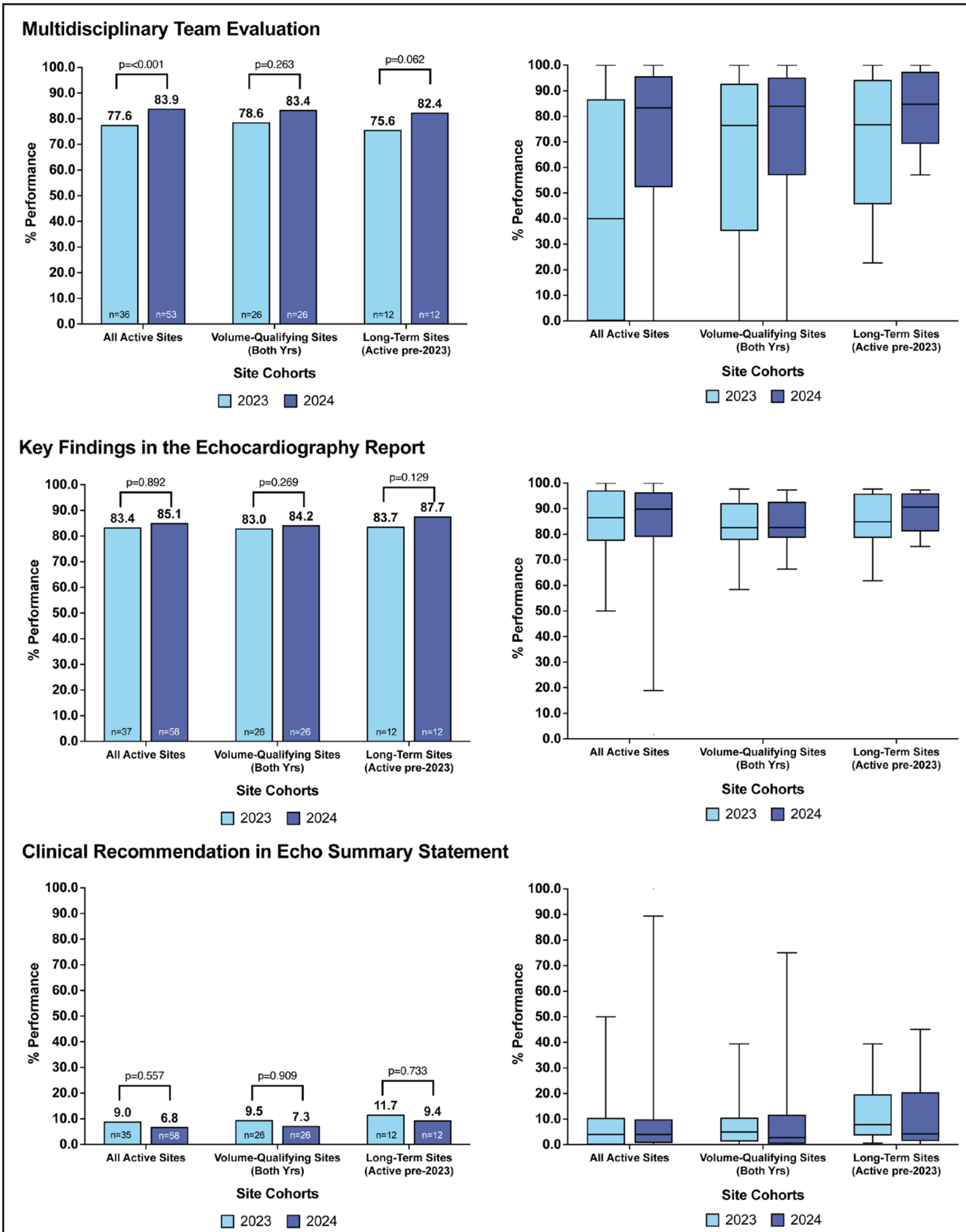

Figure 4. Multidisciplinary team evaluation, key findings in the echocardiography report, and clinical recommendation in echocardiogram (echo) summary statement (secondary quality measures in Target: Aortic Stenosis).

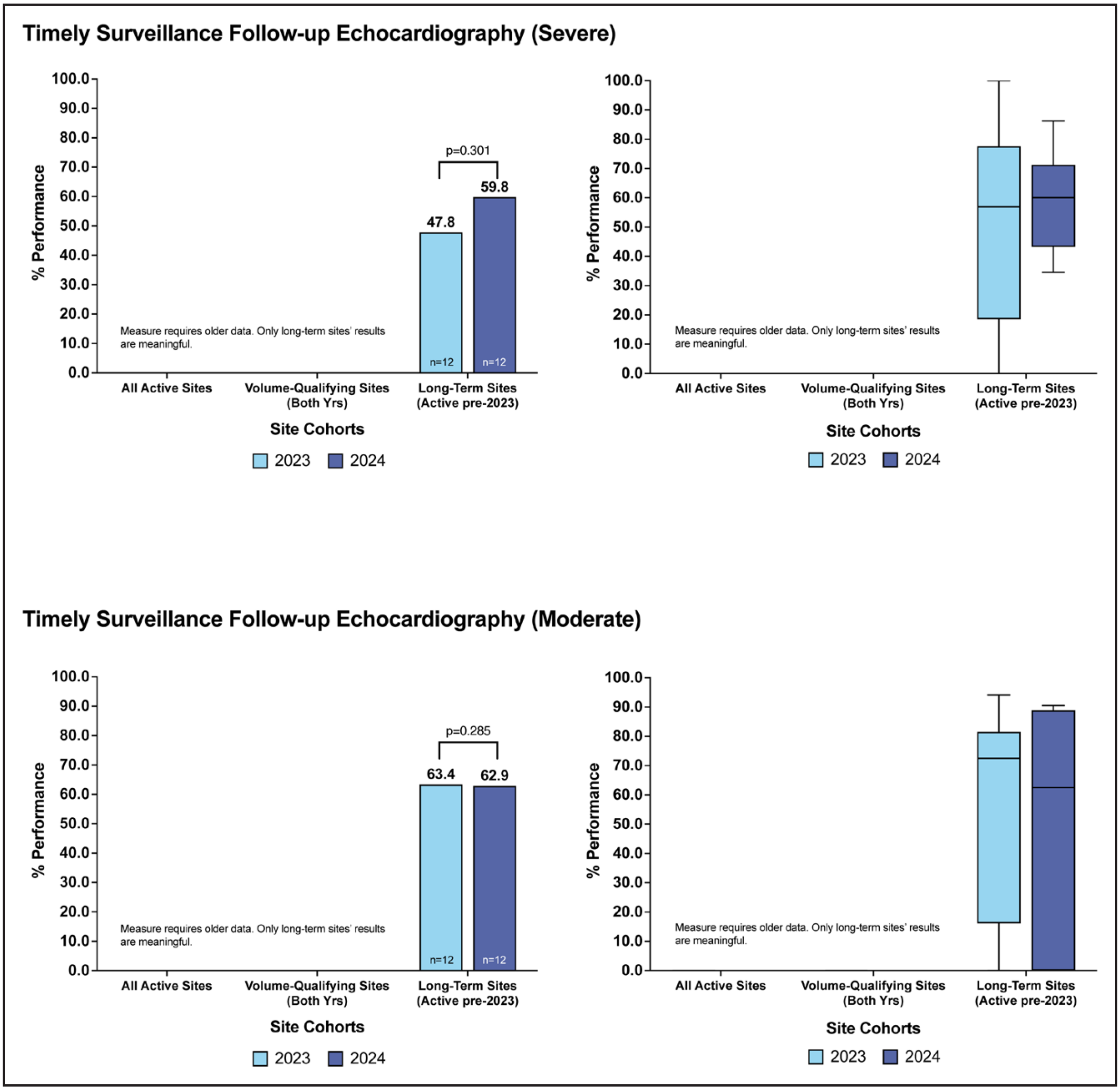

Figure 5. Timely surveillance follow-up echocardiography (secondary quality measures in Target: Aortic Stenosis).

and treatment, reflecting opportunities for improvement and sharing of best practices. Fifth, timely follow-up surveillance echocardiograms were missing 40% of the time for patients with known moderate or severe AS, which may result in delayed awareness of disease progression and late treatment.

Target: AS is structured around core quality improvement principles: standardized metric development, site-level performance feedback, iterative refinement of measures based on observed gaps, and systematic dissemination of best practices across a geographically and demographically diverse registry network. The program applies the AHA OUTPACE (optimize, understand, target, partner, advocate, communicate, and evaluate) framework, which operationalizes cardiovascular quality improvement through 7 sequential domains: optimize clinical systems, understand disparities and gaps in care, target high-risk patient populations, partner across disciplines and care settings, advocate for guideline adherence, communicate findings to providers, and evaluate outcomes. This provides a replicable structure for translating registry data into actionable practice change. Together, these elements are designed not only to measure performance but to drive the iterative, multidisciplinary improvements necessary to close the gap between guideline-recommended care and real-world management of aortic stenosis.

Although timely treatment was chosen as the primary performance measure during the pilot phase of Target: AS, there are key limitations with that measure

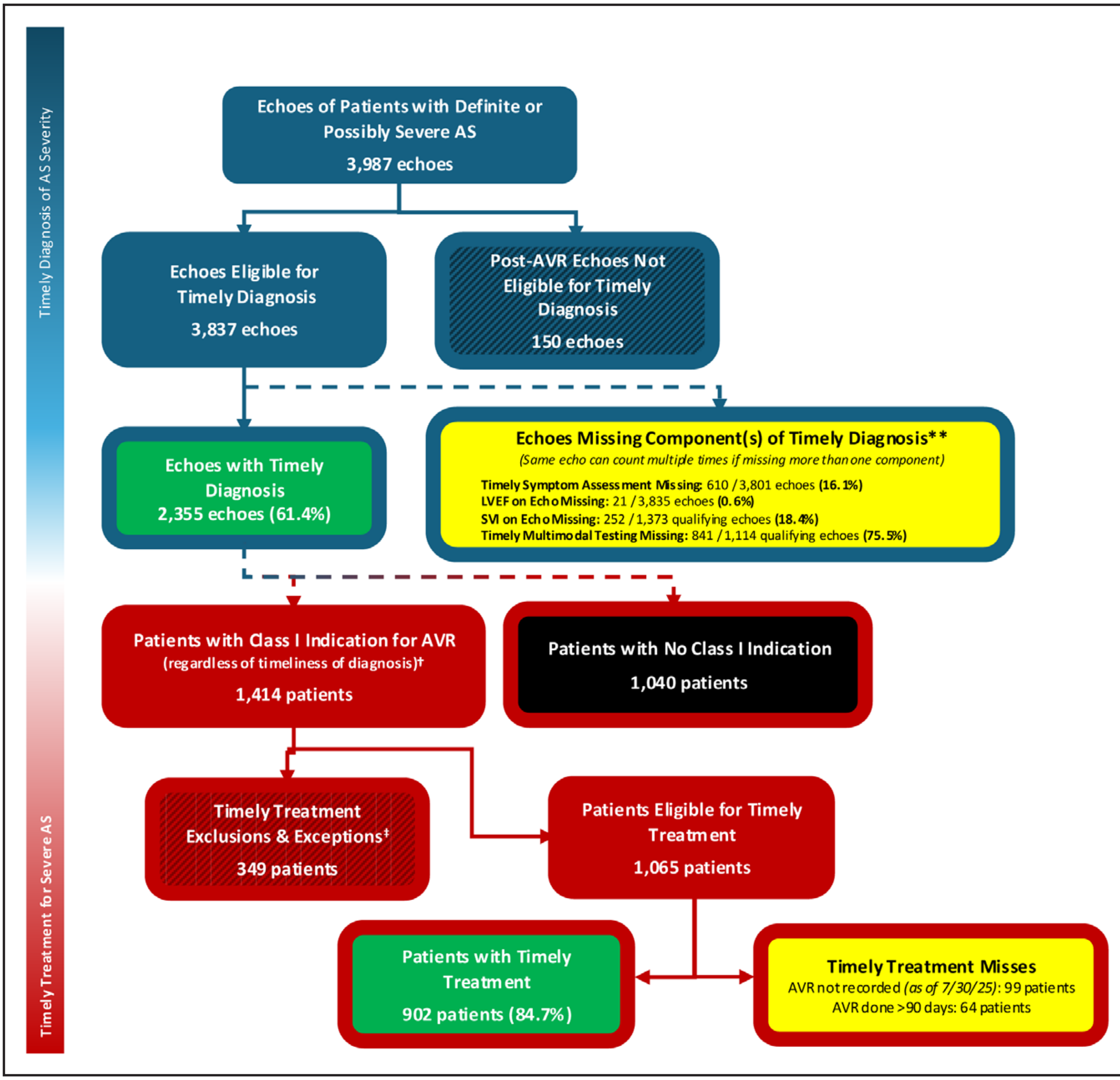

Figure 6. Timely diagnosis and timely treatment in Target: Aortic Stenosis (AS).

Results from all sites (n=54) in 2024. *Timely diagnosis exception reasons include patient died or discontinued follow-up. **Denominator values vary because each component population can qualify for an exception independently. For example, a patient who receives LVEF on echocardiogram but dies within 30 days still counts in the numerator for "LVEF on echo," whereas the same patient who did not receive multimodal testing counts as an exception for "timely multimodal testing." †Includes patients who receive confirmed diagnosis, regardless of timeliness. ‡Timely treatment reasons for exclusion and exception include previous history of AVR, patient died, patient discontinued follow-up, other acute medical issues, multidisciplinary heart team determined AVR to be futile, patient received nonprocedural treatment, patient declined AVR, or patient had qualifying reason for AVR delay. AVR indicates aortic valve replacement; echo, echocardiogram; and LVEF, left ventricular ejection fraction.

(in isolation). As is common for the AHA Get With The Guidelines programs, patients missing key data for a measure are excluded from the denominator. For Target: AS, this meant that any patient lacking data to determine whether a Class I indication for AVR existed (eg, symptom assessment, SVI, multimodal testing, etc) was excluded from the denominator, thereby artificially increasing the timely treatment rate by excluding patients lacking adequate information for accurate diagnosis. Because of this methodologic approach, the timely treatment rate reported here is substantially higher than that reported by others.[8–11] When comparing performance on timely treatment between the pilot phase and this report, it is important to point out that a refinement in the measure allowed for and characterized clinically appropriate exclusions and exceptions for this measure (eg, treatment of

AS with AVR deemed futile by the MDT, patient left the health system), which increased the performance on this measure in comparison to our previous report.[15]

To provide greater clarity on process and care gaps, a timely diagnosis measure was created, which highlighted significant gaps in diagnosis. This reflects unnecessary exposure to risk for many patients who lack important data to determine AS severity who, in fact, had severe AS (particularly if they were symptomatic). For example, Table 2 and Figure 2 show that when multimodal testing (eg, aortic valve calcium score, dobutamine stress echocardiogram) was performed in patients with discordant echocardiogram data, 57% were shown to have severe AS. Extrapolating this to the 1096 patients with discordant resting echocardiogram data who warranted—but did not receive—multimodal testing, an additional 625 patients may have had unrecognized severe AS.

The complexity of resolving discordant echocardiogram parameters may lead to heterogeneity in how patients with discordant measurements are classified on echocardiography reports.[17] Many such patients are characterized as having moderate or moderate-severe AS, which likely results in a strategy of clinical surveillance. It's possible that artificial intelligence solutions may aid in the diagnosis of severe AS and prove useful for such patients in the future.[18,19] Nevertheless, in current practice, the severity of AS may not be discerned in many cases from a resting echocardiogram alone, with the need to complete further testing to clarify severity per current guidelines.[16] Accordingly, and as recently recommended by the American Society of Echocardiography, the echocardiogram report summary for any patient with severe or potentially severe AS should indicate that significant AS is present and further evaluation may be warranted; for patients with discordant AS, further evaluation is usually warranted.[20] In Target: AS, which includes data collected before the American Society of Echocardiography publication, such a clinical recommendation was included in the summary of an echocardiography report only 10% of the time. Addressing this gap in care—failure to clarify AS severity when echocardiogram measures are discordant—will likely require a distinct clinical recommendation in the echocardiogram report conclusion acknowledging diagnostic uncertainty and prompting the ordering provider to consider further guideline recommended testing to clarify disease severity.[16,20]

Another important gap was timeliness of surveillance echocardiograms for patients with moderate and severe AS, which was missing or late in approximately 40% of patients. This may result in delayed awareness of disease progression and late treatment. This finding is consistent with recently published work from Kaiser in which they found that patients with moderate AS lacked a timely follow-up surveillance echocardiogram (within 2 years) 37.5% of the time, and patients with severe AS lacked a timely follow-up surveillance echocardiogram (within 1 year) half the time.[21] In adjusted analyses, timely follow-up echocardiography was associated with a 24% lower hazard of all-cause mortality for patients with moderate AS and a 38% lower hazard of all-cause mortality for patients with severe AS.[21] Electronic health record and artificial intelligence tools available today could help identify and address this care gap if a timely surveillance echocardiogram has not been ordered through opt-in or opt-out facilitated scheduling.

The American College of Cardiology/AHA valvular heart disease guidelines include a Class I recommendation for MDT evaluation for patients considered for AVR. For participating sites in Target: AS, this happened approximately 80% of the time (allowing for some latitude in what counted as an MDT evaluation). Ideally, this would include a clinic visit involving the patient and a heart valve team cardiologist and surgeon on the same day followed shortly thereafter by a live team discussion. It is important to note that this ideal scenario might be not realistic in many practice settings. Despite this, it is important to realize that previous data suggest that, even among cardiologists and after adjusting for patient factors, the diagnosing physician represents one of the biggest factors in determining whether a patient with severe, symptomatic AS receives an AVR.[13] This suggests that not all physicians evaluate the need or appropriateness of AVR equally, even after controlling for patient factors. Further studies examining documented reasons for nonreferral to the heart team or nonreferral for AVR have shown (1) that a substantial portion of patients have no documented reason or (2) that reasons such as dementia or frailty are often cited without robust characterization.[8,22] Taken together, these findings argue for expert evaluation by heart valve specialists to determine appropriate treatment and surveillance pathways.[10] It is important to note that the DETECT-AS (Electronic Provider Notification to Facilitate the Recognition and Management of Severe Aortic Stenosis) and ALERT (Addressing Under-Treatment and Health Equity in Aortic Stenosis and Mitral Regurgitation Using an Integrated EHR Platform) trials showed that an electronic provider notification reinforcing guideline-recommended evaluation and management, triggered by significant AS on echocardiography, improved referral and treatment rates for AS.[23–25] That a simple electronic provider notification had a measurable impact on rates of AVR and survival suggests that lack of familiarity with guideline recommendations may play a role in undertreatment of AS. Despite the improvement in AVR and survival associated with an electronic provider notification, gaps in timely treatment persisted.[23,24] As a next step, facilitated referrals (with an opt-out when treatment for AS is likely futile) to the heart valve team for patients with definite (or potentially) severe AS should be tested.

As the Target: AS initiative continues to mature and grow (91 active sites as of May 2026), there is an ambitious vision to expand this QI effort by developing a robust set of best practices that are widely and reliably deployed to optimize processes of care and patient outcomes for patients with AS. To date, there have been some efforts to identify and implement best practices (eg, checking defaults on echocardiography reporting software to ensure that stroke volume index is included; garnering stakeholder alignment to incorporate a clinical recommendation in the echocardiography report summary), but there is a need to more systematically learn and study which best practices are most associated with improving quality so that they may be disseminated and implemented across sites. Target: AS includes diverse stakeholders and seeks to build an ever-growing network of relationships toward a common goal of improved care for patients with AS. Industry partners and collaborations with other professional societies (eg, American Society of Echocardiography) have meaningfully informed the work of Target: AS and reinforced changes sought. There is a recognition that the data abstraction demands are time-consuming and expensive, and artificial intelligence tools are needed to reduce this burden. Learning collaboratives among participating sites along with clinical and quality improvement experts will continue to inform playbooks that outline implementation strategies aimed at addressing gaps in care. Research studies, both randomized and nonrandomized, will be pursued to ensure that a robust evidence base of implementation science guides changes in care pathways. Attention is also being paid to how these broad changes to processes of care will help address disparities in care related to sex and race/ethnicity. It is important to note that much work remains to be done in addressing undertreatment and delayed treatment of other forms of valvular heart diseases.

Although Target: AS represents a landmark, innovative registry that has focused on improving the diagnosis and treatment of AS, several limitations exist. This registry only includes individuals who have undergone an echocardiogram demonstrating the presence of AS, but there is a detection problem further upstream insofar as many patients with prevalent AS have not had it diagnosed yet by echocardiography due largely to the insensitivity of auscultation and patient history for detecting AS.[26,27] It is important to note that there are efforts underway to lower the burden of data submission, leveraging artificial intelligence tools to increase the number of cases submitted. Although Target: AS does convene regular site meetings to share best practices and facilitate performance improvement, substantial site-level variability in performance improvement has persisted, pointing to the need for additional care improvement techniques and tools. The echocardiography data collected in the registry is based on the report and not an analysis of the imaging quality and comprehensiveness. Undoubtedly, there is an opportunity to improve sonographer image acquisition and echocardiographer reporting, which has important implications for accurate diagnosis of AS severity; accordingly, training and education efforts of the American Society of Echocardiography and other organizations to address these gaps align with the mission of Target: AS to improve timely diagnosis and treatment of AS. Rapid implementation of the registry and onboarding of new sites as well as the narrow time range examined may have affected performance on quality metrics and analyses of change over time. The findings may lack generalizability given that earlier participating sites may differ from those joining later and those that have not joined. Last, Target: AS currently lacks long-term patient follow-up to track the impact of performance improvement efforts on patient outcomes.

CONCLUSIONS

Target: AS is a rapidly growing quality improvement initiative that was developed to identify gaps in care in the management and treatment of patients with AS. Although >80% of patients with a Class I indication for AVR were treated within 90 days, substantial care gaps exist. These include those related to timely diagnosis, mostly among patients with discordant measurements on a resting echocardiogram (AVA <1.0 cm^2 with lower gradients) who commonly lacked SVI in the echocardiography report or did not have timely multimodal testing done to clarify AS severity. In addition, timely surveillance echocardiography for moderate and severe AS was not obtained in approximately 40% of patients. Although performance improved from 2023 to 2024, ongoing care gaps and variability in performance across sites reinforce the importance of—and need to scale—this quality improvement initiative going forward to optimize patient outcomes in valvular heart disease.

ARTICLE INFORMATION

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Supplemental Material

Supplemental Methods
Tables S1–S4
Figures S1 and S2

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